exp20.py - C:\Users\pings\Desktop\exp20.py (3.14.2)

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```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
pd.set_option('display.width', 200)

df = pd.read_csv(r'C:\Users\pings\Desktop\Mall_Customers - Mall_Customers.csv')
df.info()
print(df.head())

sns.pairplot(df)
plt.show()

features = df.iloc[:, [3, 4]].values

from sklearn.cluster import KMeans
model = KMeans(n_clusters=5, n_init=10, random_state=42)
model.fit(features)

Final = df.iloc[:, [3, 4]].copy()
Final['label'] = model.predict(features)
print(Final.head())

sns.set_style("whitegrid")
sns.FacetGrid(Final, hue="label", height=8) \
    .map(plt.scatter, "Annual Income (k$)", "Spending Score (1-100)") \
    .add_legend()
plt.show()

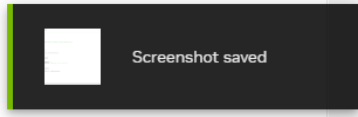
features_el = df.iloc[:, [2, 3, 4]].values
wcss = []
for i in range(1, 10):
    model = KMeans(n_clusters=i, n_init=10, random_state=42)
    model.fit(features_el)
    wcss.append(model.inertia_)
plt.plot(range(1, 10), wcss)
plt.xlabel('Number of clusters')
plt.ylabel('WCSS')
plt.show()
```

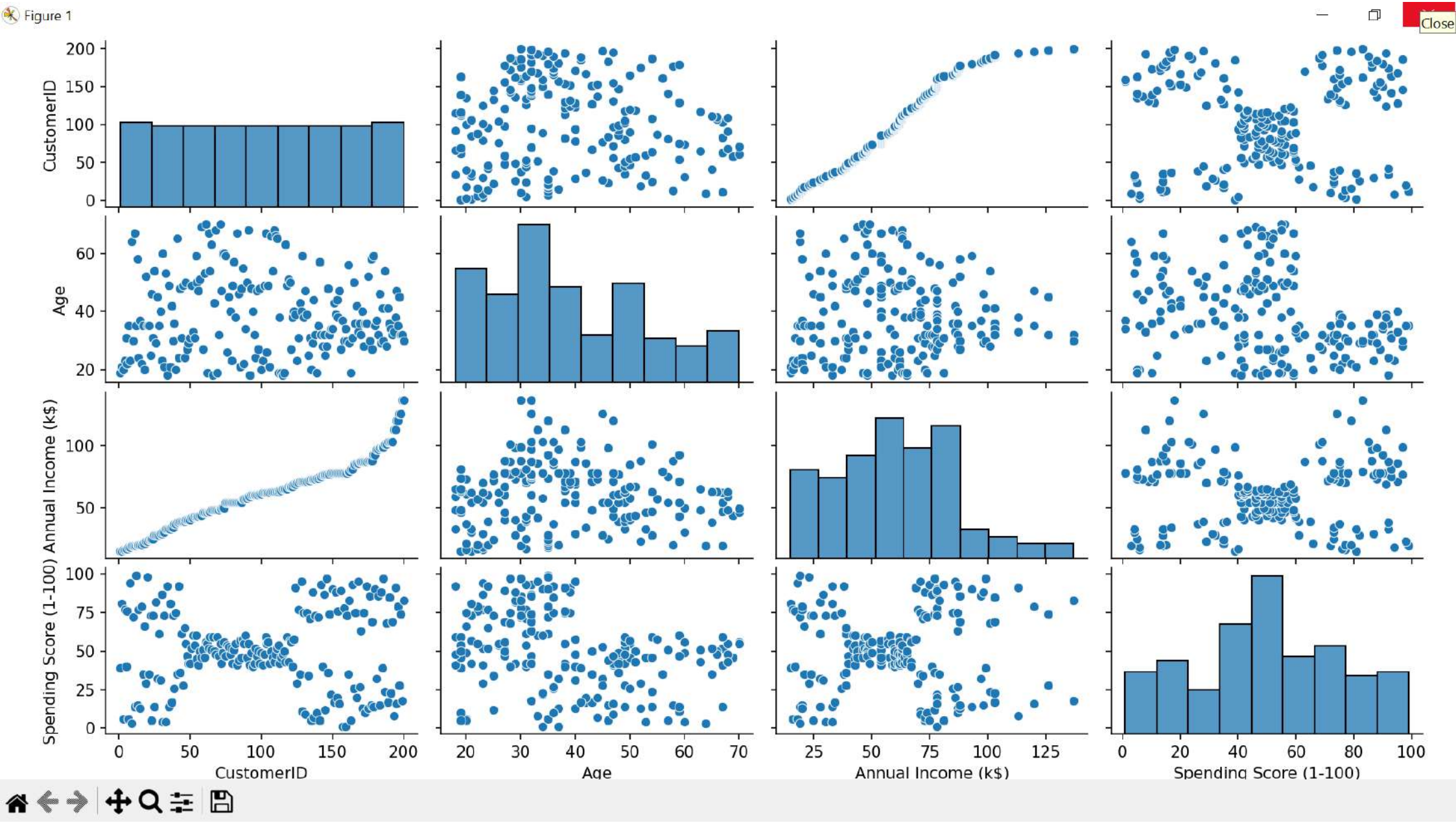
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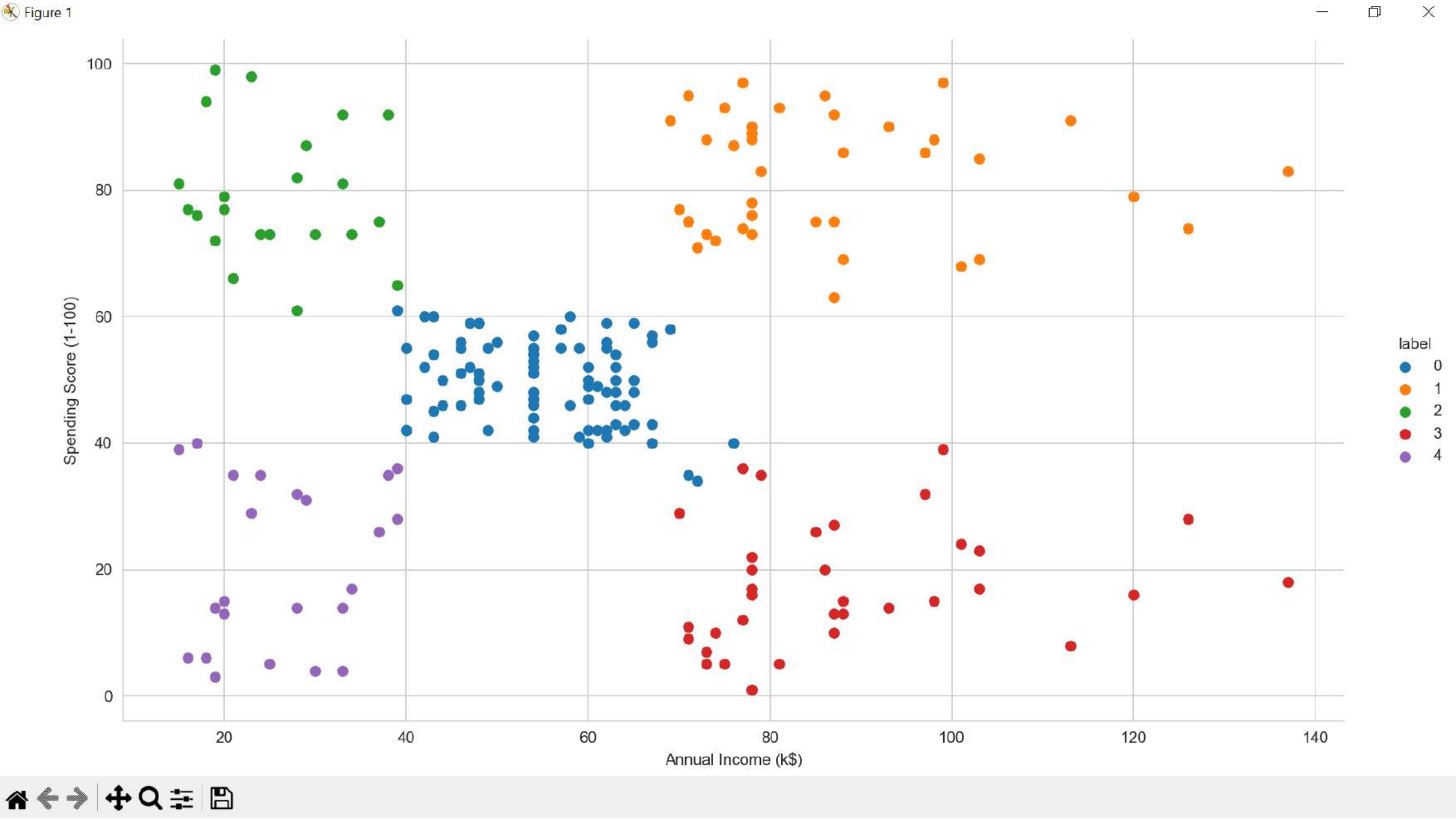
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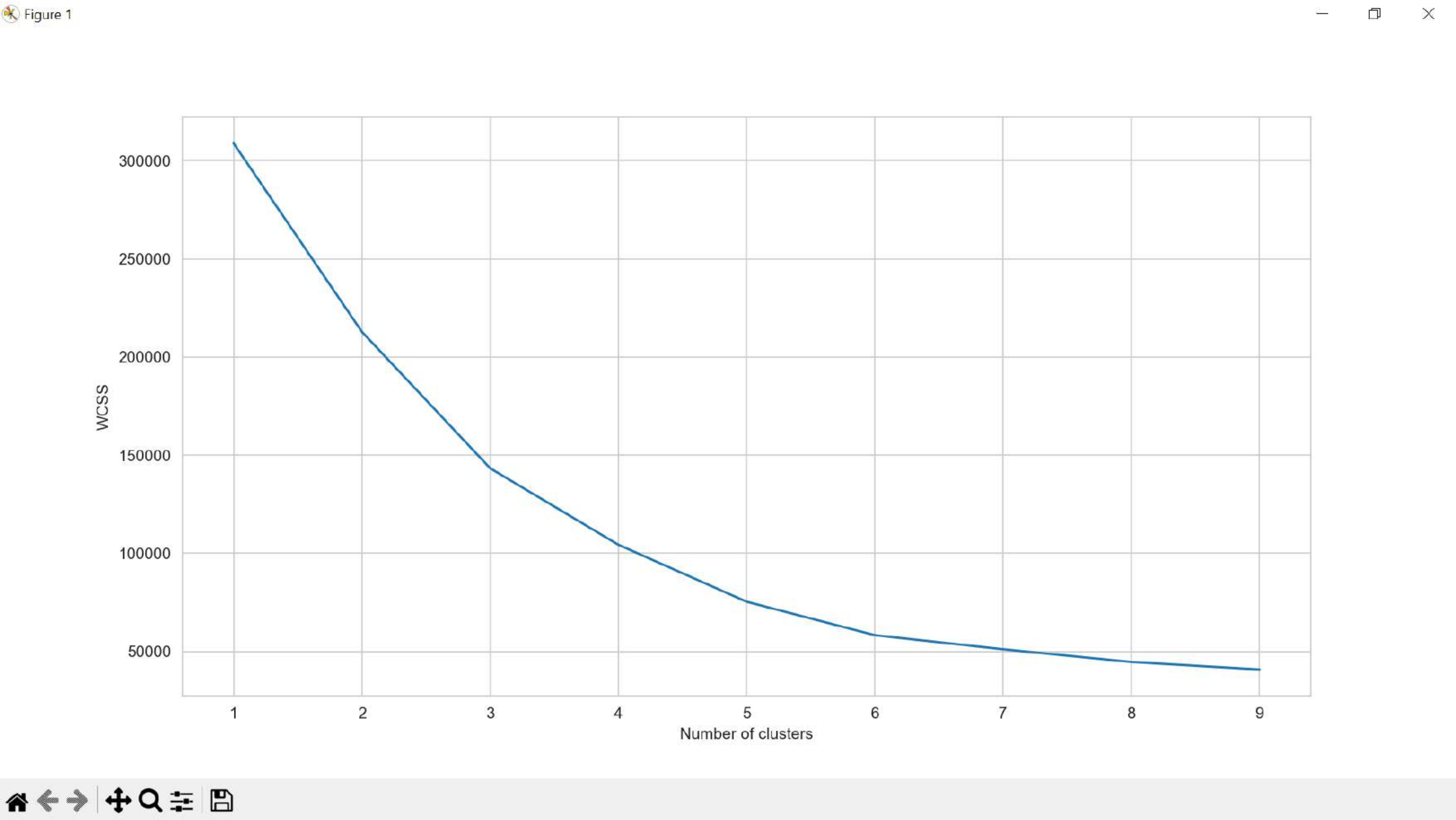
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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:\Users\pings\Desktop\exp20.py ==========
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    str
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), str(1)
memory usage: 8.9 KB
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
```




Figure 1
Close
CustomerID
Age
Annual Income (k$)
Spending Score (1-100)
0
50
100
150
200
20
30
40
50
60
70
25
50
75
100
125
0
20
40
60
80
100
0
25
50
75
100


Figure 1
Spending Score (1-100)
Annual Income (k$)
label
0
1
2
3
4
0
20
40
60
80
100
20
40
60
80
100
120
140


Figure 1
300000
250000
200000
150000
100000
50000
WCSS
1
2
3
4
5
6
7
8
9
Number of clusters